
GENERAL RELATIVITY

Tutorial problem set 13, 11.12.2020, with solutions

■ PROBLEM 34 De Sitter-Schwarzschild solution.

This problem is similar to the first problem from last week's problem set. You will solve the Einstein's equation in vacuum ($T_{\mu\nu} = 0$) for the spherically symmetric metric, only now we will assume a non-vanishing cosmological constant $\Lambda > 0$. Start with the following ansatz for the line element,

$$ds^2 = -f(t, r)dt^2 + w(t, r)dr^2 + r^2 d\Omega^2 . \quad (34.1)$$

- (a) Write down all the non-trivial components of the Einstein equation.

The non-vanishing components of the Riemann tensor for line element (34.1) are

$$R^0_{101} = \frac{1}{4f^2w} \left[2fw\partial_t^2 w - f(\partial_t w)^2 - w\partial_t f\partial_t w + f\partial_r f\partial_r w - 2wf\partial_r^2 f + w(\partial_r f)^2 \right] , \quad (34.2)$$

$$R^0_{202} = -\frac{r}{2fw}\partial_r f , \quad R^0_{303} = -\frac{r}{2fw}\sin^2 \vartheta \partial_r f , \quad (34.3)$$

$$R^0_{212} = -\frac{r}{2fw}\partial_t w , \quad R^0_{313} = -\frac{r}{2fw}\sin^2 \vartheta \partial_t w , \quad (34.4)$$

$$R^1_{212} = \frac{r}{2w^2}\partial_r w , \quad R^1_{313} = \frac{r}{2w^2}\sin^2 \vartheta \partial_r w , \quad R^2_{323} = \frac{w-1}{w}\sin^2 \vartheta , \quad (34.5)$$

and the non-vanishing components of the Ricci tensor are R_{00} , R_{11} , R_{22} , R_{33} , and $R_{01} = R_{10}$. You do not need to calculate the Ricci scalar explicitly (from the Riemann tensor), why?

Hint: examine the trace of the Einstein equation.

- (b) Argue from the tr component of the Einstein equation that w is independent of time, $w(t, r) = w(r)$.
- (c) By taking an appropriate linear combination of the remaining non-trivial components of the Einstein equation show that f must be of the form $f(t, r) = F(t)A(r)$, and $A(r) = 1/w(r)$. Further, argue that $F(t)$ can be subsumed into the redefinition of time, and needs not be considered.
- (d) Finally, show that Einstein equation implies

$$A(r) = 1 - \frac{r_S}{r} - \frac{\Lambda r^2}{3} . \quad (34.6)$$

By requiring the standard Schwarzschild limit when $\Lambda \rightarrow 0$ fix the constant r_S .

- (e) In these coordinates the horizons are located at the radii defined by $A(r) = 0$. One can determine their location by solving this condition, which is not so straightforward to solve. It is a cubic equation which can be written in a compact form as

$$x^3 - y^2(x - 1) = 0, \quad (34.7)$$

where $x = r/r_S$, $y = r_H/r_S$, and $r_H = \sqrt{3/\Lambda}$. There is a regime for y where this condition has no solutions $r > 0$, and a regime for y where it has two solutions $r > 0$. The second regime corresponds to two horizons (a black hole one and a cosmological one). In between these two regimes, there is a critical value $y = y_{cr}$ for which the horizons overlap exactly at $x = x_{cr}$. It means that (34.7) has a double root $r > 0$, and it can be factorized as

$$0 = (x - x_{cr})^2(x - \alpha), \quad (34.8)$$

where α is some constant smaller than zero, and therefore we are not interested in it. By equating (34.8) with (34.7) for $y = y_{cr}$ it is possible to determine what y_{cr} and x_{cr} are. From these values find the condition that M and Λ have to satisfy in order for the two horizons to exist.

How would you address the question of what the space-time looks like for $y < y_{cr}$?

SOLUTION: (a) We first ought to calculate the components of the Ricci tensor. For example,

$$\begin{aligned} R^0_0 &= g^{11}R^0_{101} + g^{22}R^0_{202} + g^{33}R^0_{303} \\ &= \frac{\ddot{w}}{2fw} - \frac{\dot{w}^2}{4fw^2} - \frac{\dot{f}\dot{w}}{4f^2w} + \frac{f'w'}{4fw^2} - \frac{f''}{2fw} + \frac{f'^2}{4f^2w} - \frac{f'}{rfw} \end{aligned} \quad (34.9)$$

$$R^0_1 = g^{22}R^0_{212} + g^{33}R^0_{313} = -\frac{\dot{w}}{rfw} = g^{00}R_{10} = g^{00}g_{11}R^1_0 \quad (34.10)$$

$$\begin{aligned} R^1_1 &= g^{00}R^1_{010} + g^{22}R^1_{212} + g^{33}R^1_{313} \\ &= \frac{\ddot{w}}{2fw} - \frac{\dot{w}^2}{4fw^2} - \frac{\dot{f}\dot{w}}{4f^2w} + \frac{f'w'}{4fw^2} - \frac{f''}{2fw} + \frac{f'^2}{4f^2w} + \frac{w'}{rw^2} \end{aligned} \quad (34.11)$$

$$R^2_2 = R^3_3 = g^{22}R^0_{202} + g^{22}R^1_{212} + g^{33}R^2_{323} = -\frac{f'}{2rfw} + \frac{w'}{2rw^2} + \frac{w-1}{r^2w} \quad (34.12)$$

Next, taking a trace of the Einstein's equation, $G_{\mu\nu} + \frac{\Lambda}{c^2}g_{\mu\nu} = 0$, gives $R = 4\frac{\Lambda}{c^2}$, such that the Einstein equation in vacuum can be also written as,

$$R^\mu_\nu = \frac{\Lambda}{c^2}\delta^\mu_\nu, \quad (34.13)$$

such that we do not need to calculate the components of the Ricci scalar or Einstein tensor. Inserting (34.9) into (34.13) gives the desired equations.

- (b) Since the right hand side of Eq. (34.13) is diagonal, we have,

$$R^0_1 = -\frac{\dot{w}}{rfw} = 0, \quad (34.14)$$

from which it follows $\dot{w} = 0$, and therefore $w = w(r)$.

- (c) Subtracting the 11 from the 00 Einstein's equation yields,

$$R^0_0 - R^1_1 = -\frac{f'}{rfw} - \frac{w'}{rw^2} = 0. \quad (34.15)$$

Multiplying this by $-rw$ gives,

$$\frac{f'}{f} + \frac{w'}{w} = \partial_r [\ln(fw)] = 0. \quad (34.16)$$

Integrating this yields, $fw = F(t)$, or equivalently, $f(t, r) = F(t)A(r)$, with $A(r) = 1/w(r)$.

(d) Taking account of $f = F(t)A(r)$ and $A(r) = 1/w(r)$, the 22 Einstein equation can be rewritten as (with $c = 1$),

$$-\frac{A'}{2rAw} + \frac{w}{2rw^2} + \frac{1}{r^2} \left(1 - \frac{1}{w}\right) = -\frac{A'}{r} + \frac{1}{r^2} (1 - A) = \Lambda. \quad (34.17)$$

Upon multiplying by $-r^2$, this can be rewritten as,

$$rA' + A = (rA)' = 1 - \Lambda r^2 \implies rA(r) = C + r - \frac{\Lambda r^3}{3}, \quad (34.18)$$

where C is an integration constant, which can be fixed by requiring that – in the limit when $\Lambda \rightarrow 0$ – one obtains the Schwarzschild solution, which fixes the constant to, $C = -r_S = -2GM$, such that,

$$A(r) = 1 - \frac{r_S}{r} - \frac{\Lambda r^2}{3}, \quad r_S = 2GM. \quad (34.19)$$

The unspecified function $F(t)$ contributes to the 00 component of the line element as, $A(r)F(t)dt^2$, and therefore can be absorbed into a redefinition of time, $\sqrt{F(t)}dt = d\tilde{t}$. What remains to be done is to consider the 00 Einstein's equation, which in light of the above results can be recast as,

$$R^0_0 = -\frac{A''}{2} - \frac{A'}{r} = \Lambda, \quad (34.20)$$

which is easily checked to be satisfied by $A(r)$ in Eq. (34.19).

(e) To solve this part of the problem, first note that Eq. (34.8) can be rewritten as,

$$x^3 - (\alpha + 2x_{\text{cr}})x^2 + (2\alpha + x_{\text{cr}})x_{\text{cr}}x - \alpha x_{\text{cr}}^2 = 0. \quad (34.21)$$

Comparing this with (34.7), we see that these two equations are consistent provided, $\alpha = -2x_{\text{cr}}$, $y^2 = 3x_{\text{cr}}^2$ and $y^2 = 2x_{\text{cr}}^3$. From these we conclude that, $x_{\text{cr}} = r_{\text{cr}}/r_S = 3/2$ and $y = r_H/r_S = 3\sqrt{3}/2$, or $r_{\text{cr}} = (3/2)r_S = 3GM$, $r_H = \sqrt{3/\Lambda} = 3\sqrt{3}r_S/2 = 3\sqrt{3}GM$, from where we conclude, $\Lambda = 1/(3GM)^2$.

To answer the final question, note that when $y < y_{\text{cr}}$, one can rewrite $y^2 = y_{\text{cr}}^2 - \delta y(2y_{\text{cr}} - \delta y)$ with $\delta y = y_{\text{cr}} - y > 0$, and therefore the cubic equation (34.7) becomes,

$$(x - x_{\text{cr}})^2(x + 2x_{\text{cr}}) + \delta y(2y_{\text{cr}} - \delta y)(x - 1) = 0. \quad (34.22)$$

The last term contributes negatively for $x < 1$ and positively for $x > 1$. Since the double root is at $x = x_{\text{cr}} = \sqrt{3/2} > 1$, that means that the extra contribution is positive there, and the double root will become complex for $\delta y > 0$. To see that, notice that the approximate solution of (34.22) is, $x_{\pm} \simeq x_{\text{cr}} \pm i\sqrt{\delta y(2y_{\text{cr}} - \delta y)(x_{\text{cr}} - 1)/(3x_{\text{cr}})}$. Since the remaining real root is negative, it has no physical significance, implying that no event horizons are present for $0 \leq y < y_{\text{cr}}$.

■ **PROBLEM 35** Kruskal-Szekeres coordinates.

In this problem you will show that the singularity at Schwarzschild radius, $R_S = 2MG_N$, of the Schwarzschild metric,

$$ds^2 = - \left(1 - \frac{R_S}{r}\right) dt^2 + \left(1 - \frac{R_S}{r}\right)^{-1} dr^2 + r^2 d\Omega^2 , \quad (35.1)$$

can be removed by making use of different coordinates.

We start by defining the *Kruskal-Szekeres* coordinates, for $r > R_S$:

$$R = \left(\frac{r}{R_S} - 1\right)^{1/2} \exp\left(\frac{r}{2R_S}\right) \cosh\left(\frac{t}{2R_S}\right) , \quad (35.2)$$

$$T = \left(\frac{r}{R_S} - 1\right)^{1/2} \exp\left(\frac{r}{2R_S}\right) \sinh\left(\frac{t}{2R_S}\right) , \quad (35.3)$$

and for $r < R_S$:

$$R = \left(1 - \frac{r}{R_S}\right)^{1/2} \exp\left(\frac{r}{2R_S}\right) \sinh\left(\frac{t}{2R_S}\right) , \quad (35.4)$$

$$T = \left(1 - \frac{r}{R_S}\right)^{1/2} \exp\left(\frac{r}{2R_S}\right) \cosh\left(\frac{t}{2R_S}\right) . \quad (35.5)$$

(a) Show that the metric in Kruskal-Szekeres takes the following form

$$ds^2 = -\frac{4R_S^3}{r} e^{-\frac{r}{R_S}} (dT^2 - dR^2) + r^2 d\Omega^2 . \quad (35.6)$$

(b) How is the radial null line parametrized, in terms of R and T ?

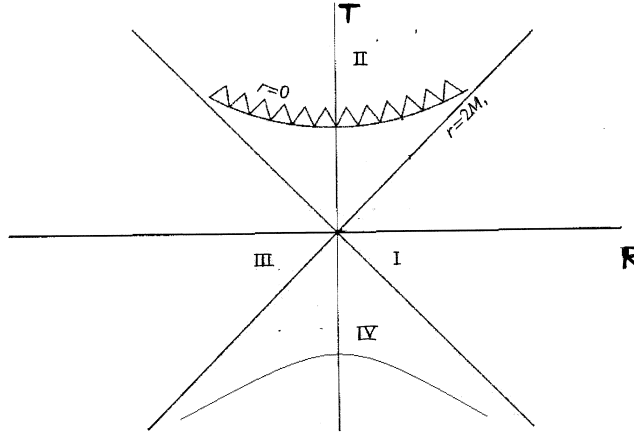


Figure 1: Kruskal diagram.

In order to get a better understanding of this space-time, we can study the diagram in Fig. 1. Every point in this diagram is supposed to represent a sphere (coordinates ϑ and φ are suppressed). The results from part (b) of this problem tell us that the light cones in coordinates T and R are as open as in SR.

- (c) What is the shape of the lines of constant r , for $r > R_S$? And for $r < R_S$? Can a massive object stay at constant r inside the horizon?
- (d) Draw the lines of constant t . What happens for $t \rightarrow \infty$?
- (e) Draw the path of an infalling observer.
- (f) Which regions are connected to region I by light signals?

Read section 5.7 from Carroll's book for a discussion of the properties of Schwarzschild black hole in Kruskal-Szekeres coordinates.

SOLUTION: (a) This problem is from section 5.7 of Carroll. To fill in the details, notice first that,

$$R^2 - T^2 = \left(\frac{r}{R_S} - 1 \right) \exp \left(\frac{r}{R_S} \right) \quad (35.7)$$

for all r , $r > R_S$ and $0 < r < R_S$. Next, let us calculate dR and dT ,

$$dR = R \frac{\frac{dr}{2R_S}}{1 - \frac{R_S}{r}} + T \frac{dt}{2R_S}, \quad (r > R_S) \quad (35.8)$$

$$dT = T \frac{\frac{dr}{2R_S}}{1 - \frac{R_S}{r}} + R \frac{dt}{2R_S}, \quad (r > R_S) \quad (35.9)$$

$$dR = R \frac{\frac{dr}{2R_S}}{1 - \frac{R_S}{r}} + T \frac{dt}{2R_S}, \quad (r < R_S) \quad (35.10)$$

$$dT = T \frac{\frac{dr}{2R_S}}{1 - \frac{R_S}{r}} + R \frac{dt}{2R_S}, \quad (r < R_S) \quad (35.11)$$

such that, for all $r > 0$, we have,

$$\begin{aligned} dR^2 - dT^2 &= (R^2 - T^2) \left[\frac{1}{\left(1 - \frac{R_S}{r}\right)^2 4R_S^2} - \frac{dt^2}{4R_S^2} \right] \\ &= \frac{r}{4R_S^3} \exp \left(\frac{r}{R_S} \right) \left[\frac{dr^2}{1 - \frac{R_S}{r}} - \left(1 - \frac{R_S}{r}\right) dt^2 \right] \end{aligned} \quad (35.12)$$

Inserting this into the Schwarzschild metric, one obtains,

$$ds^2 = \frac{4R_S^3}{r} \exp \left(-\frac{r}{R_S} \right) (-dT^2 - dR^2) + r^2(T, R) d\Omega^2, \quad (35.13)$$

which coincides with (35.6).

(b) The radial null line is parametrized by $ds^2 = 0$, or equivalently by $\frac{dR}{dT} = \pm 1$, *i.e.* the light-cones travel at 45° , which means that the Schwarzschild metric in Kruskal-Szekeres coordinates is conformal.

(c) From Eq. (35.7) we see that for all $r > 0$,

$$R^2 - T^2 = \left(\frac{r}{R_S} - 1 \right) \exp \left(\frac{r}{R_S} \right) \equiv R_0^2(r) = -T_0^2(r). \quad (35.14)$$

which represent hyperbolas. From (35.14) we see that, if $r > R_S$, $R^2 - T^2 = R_0^2(r) > 0$ are hyperbolas representing time-like curves outside the event horizon, while for $0 < r < R_S$, $R^2 - T^2 = -T_0^2(r) < 0$ are hyperbolas representing space-like curves inside the event horizon, the limiting case being $r = R_S$, at which $R^2 - T^2 = 0$ which are the two light-like curves representing the event horizons in figure 1.

(d) From Eqs. (35.3–35.5) we see that,

$$\frac{R}{T} = \coth\left(\frac{t}{2R_S}\right) \implies T = \tanh\left(\frac{t}{2R_S}\right) R, \quad (r > R_S) \quad (35.15)$$

$$\frac{R}{T} = \tanh\left(\frac{t}{2R_S}\right) \implies T = \coth\left(\frac{t}{2R_S}\right) R, \quad (r < R_S) \quad (35.16)$$

implying that, for $r > R_S$ constant t lines are straight lines with a slope less than 45° , *i.e.* they are time-like, while for $r < R_S$ constant t lines are straight lines with a slope greater than 45° , *i.e.* they are space-like. From the above analysis, we see that Schwarzschild coordinates are not suitable for a description of motion of particles inside the black hole ($r < R_S$). On the other hand, the Kruskal-Szekeres coordinates are perfectly fine there.

(e) An infalling observer starts somewhere in region I, and it will either stay in region I forever, or it will fall into the black hole (region II). If it falls into region II, it will eventually, and in finite (proper) time inevitably fall onto the singularity (denoted by the $r = 0$ ragged line in figure 1).

(f) One can reach region I from region IV (white hole). From region I one can reach region II (black hole).